COHERENCE OF THE INTERSECTION OF A QUASI-COHERENT MODULE WITH THE PUSHFORWARD OF A COHERENT SUBMODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma coherent-lemma-coherent-Noetherian (01XZ) and lemma 01Y1 (Serre subcategory). cited at proof lines 3199-3204. Characterization and stability of coherence.
- Lemma descent-lemma-finite-type-descends (05AZ). cited at proof lines 3208-3211. Coherence checked after the fpqc pullback.
- Lemma coherent-lemma-flat-base-change-cohomology (02KH). cited at proof lines 3212-3215. f^*f_*G = p_*q^*G.

Source reference. Original lemma 01XZ: sources/stacks-chow/tags.txt, source lines 2538–2538. Cited at proof lines 3199-3202 as the standing characterization of coherence; load-bearing for the reading.

Source reference. Original lemma 01Y1: sources/stacks-chow/tags.txt, source lines 2540–2540. Cited at proof lines 3200-3204 to transfer coherence along sub-quotients; load-bearing for the section case.

Source reference. Original lemma 05AZ: sources/stacks-chow/tags.txt, source lines 6867–6867. Cited at proof lines 3208-3211 to reduce coherence to that of the pullback; load-bearing.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cited at proof lines 3212-3215 to rewrite f^*f_*G as p_*q^*G; load-bearing.

Lemma 0.2. *Let $f : Y \to X$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module, let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_Y$-module, and let $\varphi : \mathcal{G} \to f^*\mathcal{F}$ be a module map. Assume*

(1) *φ is injective,*
(2) *f is quasi-compact, quasi-separated, flat, and surjective,*
(3) *X, Y are locally Noetherian, and*
(4) *$\mathcal{G}$ is a coherent $\mathcal{O}_Y$-module.*

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Then $\mathcal{F} \cap f_*\mathcal{G}$ defined as the pullback

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & f_*f^*\mathcal{F} \\ \uparrow & & \uparrow \\ \mathcal{F} \cap f_*\mathcal{G} & \longrightarrow & f_*\mathcal{G} \end{array}$$

is a coherent $\mathcal{O}_X$-module.

Proof. We will freely use the characterization of coherent modules of Cohomology of Schemes, Lemma 01XZ as well as the fact that coherent modules form a Serre subcategory of $QCoh(\mathcal{O}_X)$, see Cohomology of Schemes, Lemma 01Y1. If f has a section σ, then we see that $\mathcal{F} \cap f_*\mathcal{G}$ is contained in the image of $\sigma^*\mathcal{G} \to \sigma^*f^*\mathcal{F} = \mathcal{F}$, hence coherent. In general, to show that $\mathcal{F} \cap f_*\mathcal{G}$ is coherent, it suffices the show that $f^*(\mathcal{F} \cap f_*\mathcal{G})$ is coherent (see Descent, Lemma 05AZ). Since f is flat this is equal to $f^*\mathcal{F} \cap f^*f_*\mathcal{G}$. Since f is flat, quasi-compact, and quasi-separated we see $f^*f_*\mathcal{G} = p_*q^*\mathcal{G}$ where $p, q : Y \times_X Y \to Y$ are the projections, see Cohomology of Schemes, Lemma 02KH. Since p has a section we win. $\square$